Department of Economics

MSc Extended Essay Cover Sheet

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COURSE CODETITLE	E	C	4	2	4

TITLE	**Two Effective Channels for Central Bank Communication: Credibility and Proportion of Rational Beliefs**

WORD COUNT	3	3	8	8

Double spaced, typed. Maximum 6,000 words (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text). Examiners are recommended to penalise excessively long essays (+ 10% of word count) by adjusting the marks by 6000/x, where x is the word count. Pages are to be numbered. Binding is not necessary, however it is crucial that all pages are secure, i.e. stapled.*

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Two Effective Channels for Central Bank Communication: Credibility and Proportion of Rational Beliefs

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Abstract

Effective communication by central banks plays a crucial role in shaping economic outcomes. This essay explores the intricate relationship between communication strategies and macroeconomic stability. Our analysis focuses on a basic New Keynesian model and changes two variables of government credibility and the proportion of a household's rational beliefs. The simulation results show that The increased credibility and proportion of rational beliefs can accelerate the speed of economic recovery. However, with different types of communication, the final results may be ambiguous if we cannot quantify the impact of changes in these two variables.

1 Introduction

1.1 Main purpose of this essay

This essay wants to explore the effects of different types of communication on macroeconomic stability. To be more specific, the exogenous variables that this article wants to discuss are the central bank's credibility and the proportion of different beliefs (rational and backward-looking). This essay will change these parameters based on different ways of exogenous communication and show their effects through a basic New Keynesian model.

1.2 Importance of expected inflation and communication

The expected inflation is important as it is a key component of the current inflation, which is shown in the New Keynesian model (Clarida et al., 1999). Central banks with inflation-targeting policy can communicate with the public by publishing their forecast

about future inflation and placing the narrative in the context to convey these viewpoints and influence households' expected inflation. King et al. (2008) investigate how economic behavior can be affected by central banks' management of the expectation. Candia et al. (2020) discuss how households' inflation expectations are formed and how these expectations affect their decisions. Both researches show that the central bank's effective communication is critical in guiding expectations. Furthermore, consistent and clear messages conveyed in the communication can be crucial in stabilizing the economy through guided inflation expectations.

1.3 Importance of credibility

The central bank's credibility is important for the effectiveness of its communication. Central banks with a high level of credibility can make households more convinced by their released inflation forecasts. The ideal government with full credibility can anchor expected inflation. Woodford (2010) compares the welfare effects between commitment and discretion and shows that commitment is more beneficial. Discretion here means the central bank can not make binding commitments and inflation expectation is taken as given in the household's decision problem. On the other hand, commitment means the central bank can commit to a fully state-contingent plan for how monetary policy should be conducted. Furthermore, unconventional monetary policy such as forward guidance also depends on the central bank's credibility.

How can the central bank cultivate its credibility? Respondents of a survey conducted by Blinder (2000) think that central banks achieve high credibility by establishing records of honesty and aversion to inflation (in that order of importance), rather than controlling their discretion by commitment technologies or signing contracts compatible with incentives. Furthermore, a high level of central bank credibility can lead to virtuous circles since this allows the government to better manage inflation through the central bank's predicted inflation. On the contrary, a vicious circle can be caused because the central bank loses its control, and credibility-reducing inflation can be volatile. The experiments conducted by McMahon and Rholes (2023) find that central bank credibility depends heavily on whether past forecasts are precise.

1.4 Relation between communication and credibility

In this essay, the current level of the central bank's forecast credibility will be assumed to be affected by exogenous central bank communication. McMahon and Rholes (2023)'s experiment shows that the way of central bank communication is important. Contextualizing communication such as adding an explanation for the past forecast performance can improve the central bank's credibility significantly, especially when the past forecast performance is bad. Also, Blinder (2000)'s survey finds that factors like central bank in-

dependence and transparency are important for the central bank's credibility level. Many of these factors that determine credibility can be mentioned in communication to enhance households' confidence in the central bank.

1.5 Credibility's role in the model of this essay

The simplified timeline of the model in this essay is for household to form a prior expected inflation $\mathbb{E}^{HH}_{t,prior}\{\pi_{t+1}\}$, then the central bank releases their predicted inflation of $\mathbb{E}^{CB}_{t}\{\pi_{t+1}\}$ with communication, and finally household updates its prior expectation based on central bank's forecast and to get the post expected inflation of $\mathbb{E}^{HH}_{t,post}\{\pi_{t+1}|\mathbb{E}^{CB}_{t}\{\pi_{t+1}\}\}$ which is used in the New Keynesian model for simulation. The credibility mentioned above enters the model in the process of updating. For example, a household's updated expected inflation will be closer to the central bank's forecast when the credibility is high. This essay also assumes that a household has two types of beliefs, which are rational beliefs and backward-looking beliefs. Similar as McMahon and Rholes (2023), rational beliefs form an prior expected inflation $\mathbb{E}^{HH,RE}_{t,prior}\{\pi_{t+1}\}$ which is model consistent with noise. Backward-looking beliefs form a backward-looking prior expected inflation $\mathbb{E}^{HH,BR}_{t,prior}\{\pi_{t+1}\}$ which equals to π_{t-1}. In McMahon and Rholes (2023), they aggregate these two types of prior expected inflation ($\mathbb{E}^{HH,RE}_{t,prior}\{\pi_{t+1}\}$ and $\mathbb{E}^{HH,BR}_{t,prior}\{\pi_{t+1}\}$) first and then use this one aggregated prior expected inflation for the later updates. This essay also assumes that these two beliefs form their prior expected inflation ($\mathbb{E}^{HH,RE}_{t,prior}\{\pi_{t+1}\}$ and $\mathbb{E}^{HH,BR}_{t,prior}\{\pi_{t+1}\}$) in the first stage. But unlike McMahon and Rholes (2023), households update their prior expected inflation to post expected inflation separately based on the central bank's prediction, which leads to two post expected inflation $\mathbb{E}^{HH,RE}_{t,post}\{\pi_{t+1}\}$ and $\mathbb{E}^{HH,BR}_{t,post}\{\pi_{t+1}\}$. The model will be reintroduced in detail in the next section.

1.6 Explanations for rational and backward-looking beliefs

Besides the central bank's credibility, another parameter assumed to be affected by exogenous central bank communication is the proportion between rational and backward-looking beliefs. This proportion and the central bank's credibility mentioned above are the main focus of parameters in this essay. The backward-looking belief can be demonstrated in McMahon and Rholes (2023)'s experiment about the relation between the central bank's forecast credibility and the time when past inflation forecast errors occurred. This experiment creates three sets of data on past central bank's inflation prediction errors with the same average absolute forecast errors. However, the time of large forecast errors occur is different. The results show the evidence of recency bias which means the central bank's forecast credibility depends more on the recent predictions' accuracy. Thus, it is reasonable to add the backward-looking belief in the model.

Another important justification is that this essay assumes a single household with two types of beliefs. Therefore, it is not extreme to assume backward-looking belief will naively choose its expected inflation $\mathbb{E}_{t,prior}^{HH,BR}\{\pi_{t+1}\}$ equals to π_{t-1} without adding more information. This is because $\mathbb{E}_{t,prior}^{HH,BR}\{\pi_{t+1}\}$ is just some proportion of ideas in household's minds and will eventually combine with expected inflation from rational beliefs. This justification can also make the subsequent explanations no longer extreme.

Rational and backward-looking beliefs can be considered as two common ideas of human beings while processing information. Rational beliefs represent ideas of absorbing all the past information and making rational decisions based on all available information. As for backward-looking beliefs, they reflect ideas of focusing only on short-term information because they believe that there are unknown and frequent structural changes in the system which will make the old information useless. For example, for the economic system, backward-looking beliefs think that unknown events such as adopting a more advanced economics model or hiring some bad economic analysts happen all the time and they believe that these events can lead to structural changes in the economic system. As a result, backward-looking beliefs use the economic value from the last period as their prediction because it is the determined value with the least contamination from possible and unknown structural changes. This determined value in this essay's model is π_{t-1} because it will be assumed that the household only has the inflation part of its mind with these two beliefs.

1.7 Relation between communication and proportion of two types of beliefs

Based on the explanation from the previous paragraph, it can be easily inferred how communication affects the proportion between rational and backward-looking beliefs. The weights of the backward-looking beliefs tend to increase when the central bank's communication emphasizes the notion of structural changes. For example, during the communication, the central bank attributes the large prediction errors of the past to the misjudgment of economic analysts and ensures that more professional personnel have been hired. Under this scenario, more beliefs will change to a backward-looking style because the central bank's statement reminds them that more unknown structural changes may continue to occur in the economic system. Through this example, one can also perceive that this type of communication also affects credibility since an explanation of historical forecast performance can increase the central bank's credibility (McMahon and Rholes, 2023).

1.8 Summary of this essay

In summary, central bank communication is a powerful tool and has become increasingly important in monetary policy over the last few decades (Blinder et al., 2008). This essay assumes that exogenous central bank communication can influence the central bank's credibility perceived by backward-looking beliefs and the proportion of rational and backward-looking beliefs. By changing these two variables and doing the simulations, this essay finds that the increased credibility and proportion of rational beliefs can accelerate the speed of economic recovery. However, with different types of communication, the final results may be ambiguous if we cannot quantify the impact of changes in these two variables.

The rest of the paper is organized in the following way: Section 2 introduces the model used in this essay. Section 3 shows the results of different simulations. Section 4 discusses some possible future studies, and Section 5 concludes.

2 Model introduction

2.1 Main setup

The setup in this essay is based on McMahon and Rholes (2023) with modification and simplification. The household in this model has two types of beliefs, rational and backward-looking, and these two beliefs have different ways of forming the expected inflation and adjusting the expected inflation. This essay also assumes that the household only has the inflation part of its mind with these two beliefs, which means that the concepts of these two beliefs will not be applied to predicting future output gaps in the later New Keynesian model. Similar to what is mentioned in the first section, the complete timeline in the model is that both types of beliefs form their prior expected inflation first. In the second stage, the central bank posts its inflation forecast with communication. Based on the communication, the backward-looking beliefs may change their perception of the central bank's credibility Γ_t, and the household may change the proportion between these two types of beliefs ξ_t in its inflation minds. In the last stage, both beliefs in households' updated inflation minds will adjust their prior expected inflation to their post expected inflation.

At the beginning of time t, $\Gamma_{prior,t}$ presents the central bank's prior credibility perceived by the backward-looking beliefs. Rational beliefs accounts for $\xi_{t,prior}$ percent of the household's inflation mind, and the remaining parts belong to backward-looking beliefs. In the first stage, rational beliefs will form their prior expected inflation $\mathbb{E}_{t,prior}^{HH,RE}\{\pi_{t+1}\}$

which is model consistent with noise:

$$\mathbb{E}_{t,prior}^{HH,RE}\{\pi_{t+1}\} = \pi^{RE} + \epsilon_{HH}, \qquad \epsilon_{HH} \sim \mathcal{N}(0, \frac{1}{\alpha})$$

α represents the accuracy of rational beliefs' prior prediction, and the higher α is, the higher the accuracy of prediction. The *HH* in the formula refers to the household. Backward-looking beliefs will form their prior inflation expectation as $\mathbb{E}_{t,prior}^{HH,BR}\{\pi_{t+1}\}$:

$$\mathbb{E}_{t,prior}^{HH,BR}\{\pi_{t+1}\} = \pi_{t-1}$$

In the second stage, the central bank publishes its inflation forecast $\mathbb{E}_{t}^{CB}\{\pi_{t+1}\}$ which is also model consistent but noisy:

$$\mathbb{E}_{t}^{CB}\{\pi_{t+1}\} = \pi^{RE} + \epsilon^{CB}, \qquad \epsilon_{CB} \sim \mathcal{N}(0, \frac{1}{\beta})$$

β represents the accuracy of the central bank's forecast, and the higher β is, the higher the accuracy of the forecast. The *CB* in the formula refers to the central bank. The central bank can also communicate with the household at this stage and the backward-looking beliefs may adjust their perception of the central bank's credibility from $\Gamma_{t,prior}$ to $\Gamma_{t,post}$. Also, households can update the prior proportion of two beliefs ($\xi_{t,prior}$ and 1 - $\xi_{t,prior}$) to the post proportion ($\xi_{t,post}$ and 1 - $\xi_{t,post}$), and the changed beliefs will repeat the first stage and form their new prior inflation expectation. Further, we have:

$$\Gamma_{t-1,post} = \Gamma_{t,prior}$$
$$\xi_{t-1,post} = \xi_{t,prior}$$

In the third stage, based on central bank's forecast, two types of beliefs in updated inflation minds adjust their expected inflation to $\mathbb{E}_{t,post}^{HH,RE}\{\pi_{t+1}|\mathbb{E}_{t}^{CB}\{\pi_{t+1}\}\}$ and $\mathbb{E}_{t,post}^{HH,BR}\{\pi_{t+1}|\mathbb{E}_{t}^{CB}\{\pi_{t+1}\}\}$. Rational beliefs will adjust their post expected inflation to a linear combination of their prior expected inflation and the central bank's forecast by the Bayesian rule (McMahon and Rholes, 2023):

$$\mathbb{E}_{t,post}^{HH,RE}\{\pi_{t+1}|\mathbb{E}_{t}^{CB}\{\pi_{t+1}\}\} = \frac{\tilde{\alpha}\mathbb{E}_{t,prior}^{HH,RE}\{\pi_{t+1}\} + \tilde{\beta}\mathbb{E}_{t}^{CB}\{\pi_{t+1}\}}{\tilde{\alpha} + \tilde{\beta}}$$

$\tilde{\alpha}$ and $\tilde{\beta}$ are rational belief's estimator of α and β, which is assumed to be consistent in this essay. The backward-looking beliefs will also adjust their post expected inflation by a linear combination of its prior expected inflation and the central bank's prediction. The weights of the linear combination depend on their perception of the central bank's

credibility of forecasting:

$$\mathbb{E}_{t,post}^{HH,BR}\{\pi_{t+1}|\mathbb{E}_t^{CB}\{\pi_{t+1}\}\}=\Gamma_{t,post}\mathbb{E}_t^{CB}\{\pi_{t+1}\}+(1-\Gamma_{t,post})\mathbb{E}_{t,prior}^{HH,BR}\{\pi_{t+1}\}$$

As explained above, Γ represents the level of government credibility perceived by backward-looking beliefs, which is assumed to be exogenous in this essay, and the higher Γ is, the higher the perceived credibility.

Eventually, we get the household's aggregate post expected inflation:

$$\mathbb{E}_{t,post}^{HH}\{\pi_{t+1}|\mathbb{E}_t^{CB}\{\pi_{t+1}\}\}=\xi_{t,post}\mathbb{E}_{t,post}^{HH,RE}\{\pi_{t+1}|\mathbb{E}_t^{CB}\{\pi_{t+1}\}\}+(1-\xi_{t,post})\mathbb{E}_{t,post}^{HH,BR}\{\pi_{t+1}|\mathbb{E}_t^{CB}\{\pi_{t+1}\}\}$$

This essay does the simulation based on Walsh (2017)'s simple 3-equation New Keynesian model linearized around a zero-inflation steady-state:

$$\begin{aligned}
y_t &= \mathbb{E}_{i,t}\{y_{t+1}\}-\sigma^{-1}(i_t-\mathbb{E}_{t,post}^{HH}\{\pi_{t+1}|\mathbb{E}_t^{CB}\{\pi_{t+1}\}\})+g_t\\
\pi_t &= \beta\mathbb{E}_{t,post}^{HH}\{\pi_{t+1}|\mathbb{E}_t^{CB}\{\pi_{t+1}\}\}+\kappa y_t+u_t\\
i_t &= \phi_y y_t+\phi_\pi\pi_t+v_t
\end{aligned}$$

where

$$\begin{aligned}
g_t &= \rho_g g_{t-1}+\epsilon_t^g\\
u_t &= \rho_u u_{t-1}+\epsilon_t^u\\
v_t &= \rho_v v_{t-1}+\epsilon_t^v
\end{aligned}$$

2.2 Simplifications of the model during the simulation

First, as mentioned above, backward-looking beliefs' perception of the central bank's credibility Γ, and the proportion between rational and backward-looking beliefs (ξ and $1-\xi$) are assumed to be exogenous. Rational belief's consistent estimators $\tilde{\alpha}$ and $\tilde{\beta}$ are also assumed to be exogenous. This essay specifies the following one-time communication rule, starting from a zero-inflation steady-state at time 0, the central bank only does the contextual communication once at time 0. This one-time communication rule implies that the central bank only publishes its forecast inflation after time 0 without doing contextual communication. Therefore, the Γ and ξ are fixed and equal to $\Gamma_{t=0,post}$ and $\xi_{t=0,post}$ after time 0. In the rest of the essay, Γ and ξ will be used to represent these two fixed numbers.

3 Simulations and results

3.1 Parameter values

Starting from a zero-inflation steady-state, this essay will show the impulse response of a one percentage exogenous cost-push shock occurring at time 0. For rational belief's consistent estimator, this essay assumes that $\tilde{\alpha}$ is some finite number and $\tilde{\beta} = \infty \ \forall t$, which implies that the central bank is fully credible for rational beliefs. Thus, $\mathbb{E}_{t,post}^{HH,RE}\{\pi_{t+1}|\mathbb{E}_t^{CB}\{\pi_{t+1}\}\} = \mathbb{E}_t^{CB}\{\pi_{t+1}\}$. The rest of the parameter values are shown in Table 1 from McMahon and Rholes (2023).

Parameter Values							
σ	β	κ	ϕ_y	ϕ_π	ρ_g	ρ_u	ρ_v
1	0.99	0.104	0	1.5	0.5	0.5	0

Table 1: Parameter values for simulation

3.2 Effects from changing backward-looking beliefs' perception of the central bank credibility Γ

Firstly, with fixed ξ, this essay will show the effects of Γ which is backward-looking beliefs' perception of the central bank's credibility. The central bank can use contextualizing communication such as explaining its results or comparing its performance with peer forecasting institutions to improve its credibility significantly (McMahon and Rholes, 2023). Thus, sincere communication may increase Γ. This essay assumes the one-time communication rule that the central bank will only communicate once at time 0, and this communication will permanently change Γ. In this section, this essay assumes that $\xi = 0.4$ and changes Γ from 1 to 0 with an interval of 0.2. The results are shown in Figure 1. The speed of recovery increases with higher central bank credibility perceived by backward-looking beliefs. This increase in Γ enhances the central bank's ability to influence the expected inflation of household's backward-looking beliefs, which leads to less inflation persistence following the cost-push shock, compared to the situation where the central bank can not maintain a high level of its forecast credibility.

3.3 Effects from changing proportion between rational and backward-looking beliefs (ξ and $1 - \xi$)

Secondly, with fixed Γ, this essay will show the effects of ξ which is the proportion between rational and backward-looking beliefs. As explained above, rational beliefs make decisions based on all available information. In contrast, backward-looking beliefs focus solely on short-term information because they think that there are frequent and unknown

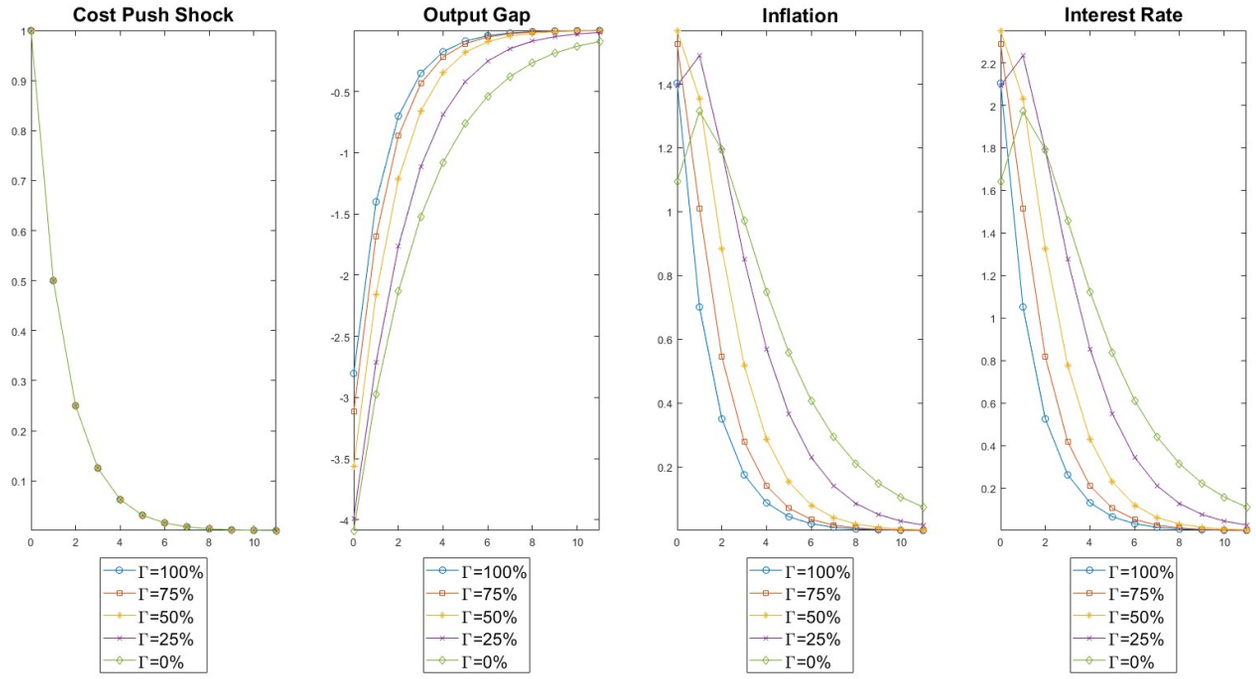

Figure 1: Effects of changing Γ

structural changes in the system, making old information useless. Therefore, contextual communication emphasizing the notion of structural change may cause ξ to update. For example, the central bank might attribute its poor performance to staff errors and promises that these individuals have been dismissed, with more professional economic analysts hired. This way of communication provides evidence for the structural changes in the economic system and may increase the share of backward-looking beliefs in the household's mind. With the same one-time communication rule, this essay assumes that $\Gamma = 0.7$ and changes ξ from 1 to 0 with an interval of 0.2 in this section. The results are shown in Figure 2. The speed of recovery increases with the proportion of rational beliefs. Due to the simplification mentioned earlier that $\mathbb{E}_{t,post}^{HH,RE}\{\pi_{t+1}|\mathbb{E}_t^{CB}\{\pi_{t+1}\}\} = \mathbb{E}_t^{CB}\{\pi_{t+1}\}$, this increase in ξ also improve the central bank's ability to influence household's inflation expectation because ξ percentage of household's beliefs consider the central bank as completely credible. So, there will be less inflation persistence following the cost-push shock, relative to the world in which the proportion of rational beliefs is low.

3.4 Combined effects from changing both Γ and ξ

Thirdly, this essay will show the effects of changing both backward-looking beliefs' perception of the central bank's credibility Γ and rational beliefs' proportion ξ. This is also the main focus of this essay, which is the effects of different ways of communication.

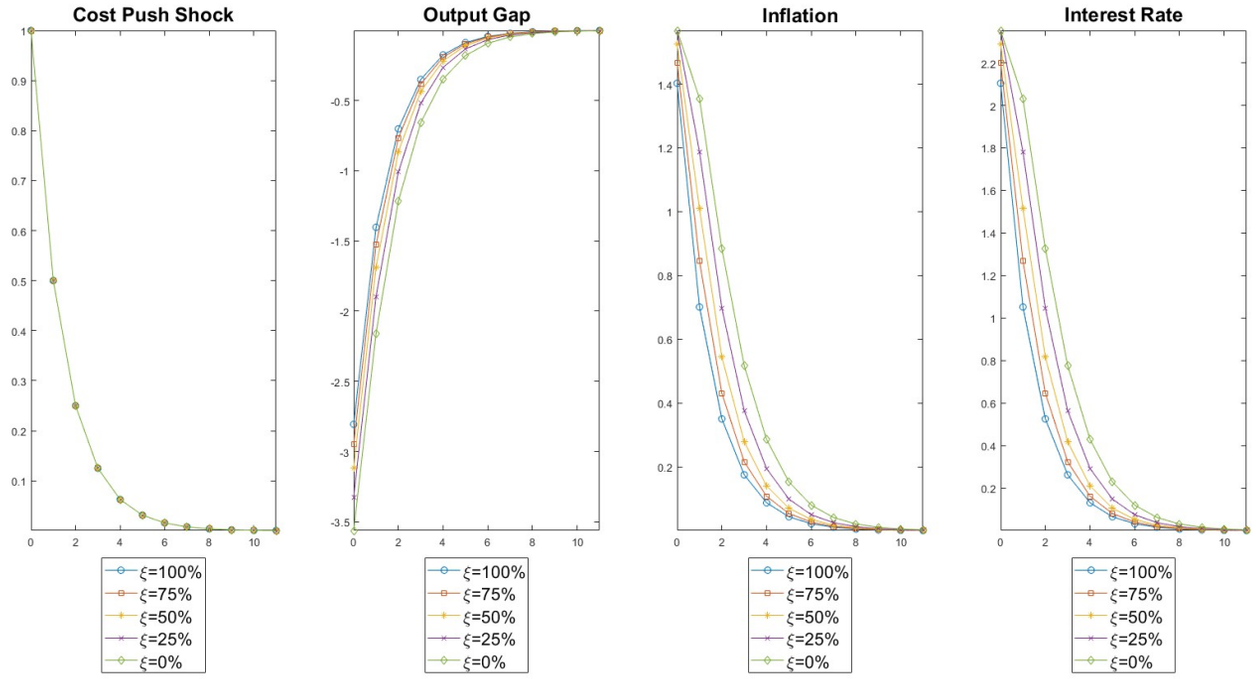

Figure 2: Effects of changing ξ

Following the discussion above, situations where both parameters change may happen in many scenarios. For example, when a central bank has done a good job at forecasting for several periods, the level of its credibility is high, it would advertise its work for political purposes such as the next election. For example, during contextual communication, the government may praise itself by saying that it constantly updates its prediction model. This way of communication can continue to increase credibility which is reflected in a higher Γ. On the other hand, by emphasizing technology that is constantly updated, some parts of rational beliefs may switch into backward-looking beliefs because they realize the unknown structural changes that may be happening all the time. This force can reduce ξ. With increasing Γ and decreasing ξ, based on the simulation from the last two sections, the aggregate effects of this type of communication are predicted to be ambiguous. Furthermore, since the level of credibility is already high given their good prediction in the past, the marginal increase in Γ might be small and the effect from decreasing ξ may be more likely to dominate. Alternatively, in the situation also mentioned in the last section, the central bank does poorly in prediction and blames their employees for this bad performance. If the households believe in the central bank's excuse, the credibility may increase and Γ is higher, but the decreased ξ may also make the final results ambiguous. With the same one-time communication rule, this essay compares the following 4 cases:

1. $\Gamma = 0.5$ and $\xi = 0.5$ used as a benchmark for comparison.

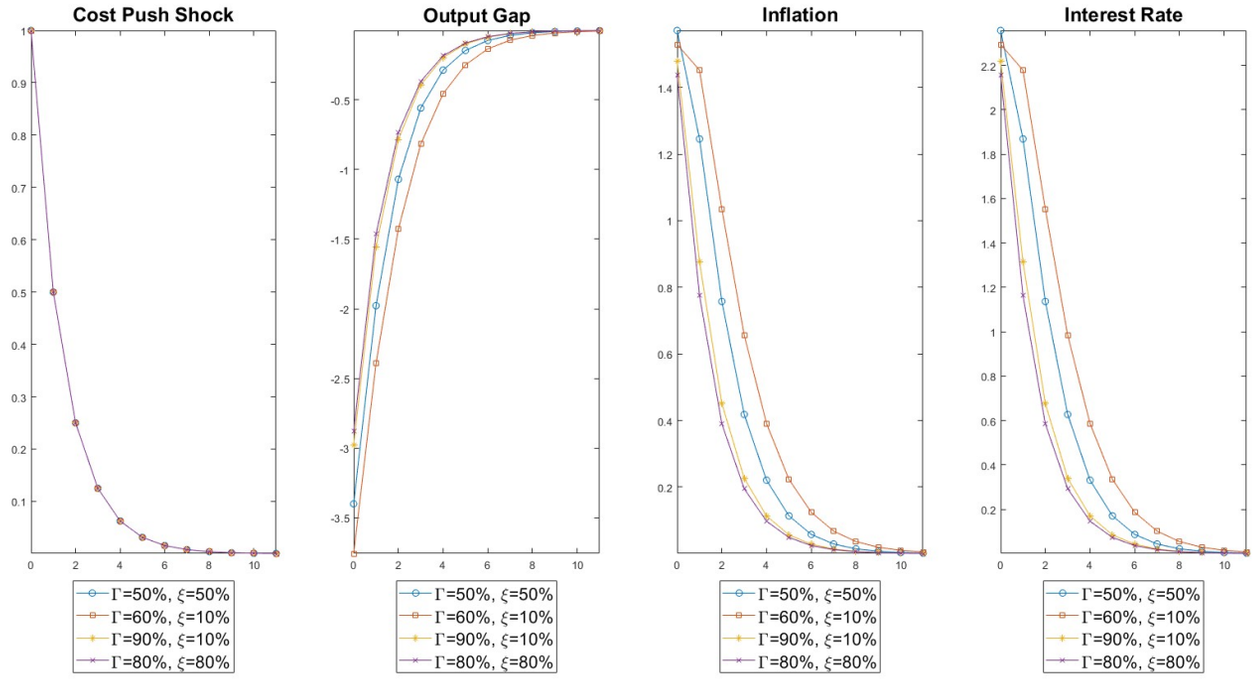

Figure 3: Effects of changing Γ and ξ

2. $\Gamma = 0.6$ and $\xi = 0.1$ so that Γ increases a little but ξ decreases a lot comparing to benchmark

3. $\Gamma = 0.9$ and $\xi = 0.1$ so that Γ increases a lot and ξ decreases a lot comparing to benchmark

4. $\Gamma = 0.8$ and $\xi = 0.8$ so that both Γ and ξ increase comparing to benchmark

The results are shown in Figure 3. When comparing case 1 and case 2, we observe that if the central bank's communication leads to a small increase in Γ and a large decrease in ξ, the speed of recovery will decrease. However, by comparing case 1 and case 3, it is find that if the increase in Γ is sufficiently high, the recovery speed will actually increase. The comparison between case 1 and case 4 shows that communication is most powerful if it increases both Γ and ξ.

3.5 Some real-world implications

Although this essay doesn't conduct empirical studies to investigate the relative power between Γ and ξ, we can still see from the simulation that some types of communication with increasing credibility Γ and decreasing rational beliefs' proportion ξ may lead to a slower recovery speed. Therefore, to enhance the impact of communication, it's essential that contextual communication aimed at increasing Γ also takes care not to inadvertently

reduce ξ. Balancing these factors ensures a more effective outcome. For example, when the central bank has done a bad job at forecasting inflation, it should not emphasize the notion of structural changes such as blaming its bad performance on the last manager who has been fired. Avoiding the concept of structural change helps maintain the level of ξ. They should also carefully explain their mistakes and sincerely promise they will do better in the future to increase their credibility level Γ.

4 Future studies

For the model part, in this essay, rational beliefs' estimator of the central bank's forecast precision $\tilde{\beta}$ and backward-looking beliefs' perception of the central bank's credibility Γ are assumed to be exogenous. However, they may be endogenous and correlated with the central bank's past prediction performance (McMahon and Rholes, 2023). Further studies can enhance the model by making some exogenous parameters endogenous. In the simulation, by the one-time communication assumption, this essay simplifies the process and makes Γ, $\tilde{\beta}$ and ξ static. Nevertheless, there can be dynamic relations within these variables.

For the empirical part, this essay mainly discusses the effects of changing Γ and ξ through different ways of communication. The effects of these two parameters are shown in the simulation. However, the relations between these two parameters and the communication are not fully discussed. Parts of the relations are shown in McMahon and Rholes (2023)'s experiment that contextual communication can increase credibility Γ. Therefore, we still need empirical evidence showing the relationship between different ways of communication and ξ, the proportion of rational beliefs. For example, future studies can be conducted to show whether communications emphasizing possible structural changes can indeed decrease ξ in real life.

5 Conclusion

This essay has explored the critical role of central bank communication in shaping macroeconomic stability. By analyzing the possible effects of different communication strategies through two channels, credibility and proportion of rational beliefs, we have gained valuable insights into how these two variables impact economic outcomes. The increased credibility and proportion of rational beliefs can accelerate the speed of economic recovery. However, with different types of communication, the final results may be ambiguous if we cannot quantify the impact of changes in these two variables. Our findings also offer practical implications. During the central bank's communication, it should focus on increasing its credibility while avoiding decreasing the proportion of the household's rational beliefs. The central bank can accomplish this through sincere explanation without

mentioning the notion related to structural change. Future studies can add more dynamic aspects to the simulation. Moreover, research should be conducted to explore the empirical evidence that shows the relationship between different ways of communication and the proportion of rational beliefs.

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